MILITARY SPECIFICATION

INDICATOR, BEARING-DISTANCE-HEADING ID-1103/ARN

1. SCOPE

1.1 This specification covers the requirements for one type of bearing-distance-heading indicator, designated ID-1103/ARN.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall

Military

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-C-5824	Course Indicator ID-250A/ARN
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-16892	Synchros, 400 Cycles
MIL-P-26514	Polyurethane Foam, Rigid or Elastic, for Packaging
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge and Cover

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STANDARDS

Federal

FED-STD-1	Standard for Laboratory Atmospheric Conditions for Testing
FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment
FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-710	Synchro, 60 and 400 Cycle
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-889	Dissimilar Metals
MS27572	Case, Instrument, Flange-mounted, Aircraft
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

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3.3 Materials. Materials shall conform to the applicable specifications as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.2 Corrosive fumes. The materials, as installed in the indicator and under the service conditions specified herein, shall not liberate deleterious fumes.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be encountered in storage or normal service.

3.3.3.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889, requirement 16.

3.3.4 Magnetic materials. Magnetic materials shall not be used for parts of the indicator, except where absolutely essential.

3.4 Design and construction

3.4.1 Design

3.4.1.1 Reliability

3.4.1.1.1 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-781.

3.4.1.1.2 Mean-time-between-failures (MTBF). The indicator shall have a minimum acceptable MTBF of 500 hours at a 90 percent confidence level.

3.4.1.1.3 Longevity. The indicator shall have a minimum operational life (equipment longevity as defined in MIL-STD-781) of 2,500 hours before wearout failures occur or the equipment consistently fails to meet the specified MTBF requirements.

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3.4.1.2 Maintainability.

3.4.1.2.1 Qualitative requirements. The following shall be considered in the design of the indicator:

a. Minimization of complexity of maintainance tasks (for example, calibration, adjustment, inspection, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performances

c. Minimum number and types of tools and test equipment (special and standard) needed to perform maintenance

d. Optimum accessibility in all components requiring maintenance inspection, removal, or replacement.

3.4.1.2.2 Quantitative requirements. The indicator shall have a mean corrective maintenance downtime (M_{c_t}) of 100 minutes, not including the time required to unseal and seal the indicator. The maximum corrective maintenance downtime ($M_{MAX_{ct}}$) shall not exceed 300 minutes not including the unsealing and sealing time.

3.4.2 Construction. The indicator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.2.1 The indicator shall be so constructed that overhaul, including replacement of worn parts, adjustment, balance, and calibration may be accomplished by instrument and electronic repairmen provided with simplified tools and equipment, and trained in the repair of electronic equipment.

3.4.3 Case. The outline dimensions of the indicator case shall conform to MS27572 for the 3-inch size. Dimension L shall not exceed 7.5 inches, and dimension G shall be 0.470 to 0.545 inch. The case shall provide a hermetically sealed enclosure for the instrument mechanism. The design shall be such that the mechanism may be removed from the case, replaced, and the case resealed without the use of special tools or equipment, except as may be approved by the procuring activity. Electrical components of the indicator, which are in themselves sealed or otherwise suitable for replacement during the service life of the indicator, may be placed external to the sealed case under a suitable cover.

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3.4.3.1 Body and mounting flanges. The body and mounting flanges of the case shall be made of lightweight nonmagnetic metal, shall be uniform in texture, and shall have a smooth surface. (see 3.9.3.)

3.4.3.2 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall not contain more than 0.006 milligram of water vapor per liter (dewpoint -65°) at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere (12 to 14.7 psi).

3.4.3.2.1 Leakage. The indicator case shall be so sealed that at a pressure differential of 12 to 14.7 psi, the leak rate will not exceed 0.1 micron cubic foot per hour.

3.4.4 Electronic components. Electronic components and the application thereof shall be in accordance with MIL-STD-883 and the class 2 requirements of MIL-E-5400.

3.4.4.1 Electrical power requirements. The equipment shall be designed to meet all applicable requirements of MIL-STD-704. The indicator shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified herein, except that the normal input voltage shall be 26V ac. The power required shall not exceed the following:

a. A-C- power (single phase)

26V (constant frequency)

Operating voltage limits . . . . . 24 to 28V

Operating frequency limits . . . . . 380 to 420 Hz

Power . . . . . 45 va

b. D-C power (28-V system)

Operating voltage limits . . . . . 25 to 29V

Power 4.5w

3.4.4.1.1 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 for class A1 equipment.

3.4.4.2 Undervoltage protection. The indicator shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

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3.4.4.3 Electrical receptacle. All indicator circuits shall be wired to a hermetically sealed receptacle mounted on the indicator. The receptacle shall be a Cannon No. KH 30-21-30PN, or an approved equivalent.

3.4.4.4 Electrical connections. The following connections shall be used on the receptacle mounted on the indicator:

A. Lead Y of compass card synchro control transformer

B. Lead X of compass card synchro control transformer

C. Lead Z of compass card synchro control transformer

D. Lead C of compass card synchro control transformer and internal ground connection

E. Leads Z and C of single-bar bearing pointer synchro receiver

F. Lead Y of single-bar bearing pointer synchro receiver

H. Lead X of single-bar bearing pointer synchro receiver

J. Lead H of single-bar bearing pointer synchro receiver

K. 26-V, 400-Hz servo motor and rate generator excitation

L. Lead H of all distance synchro receivers

M. Lead X of double-bar bearing pointer synchro receiver

N. Lead Y of double-bar bearing pointer synchro receiver

P. Lead Y of hundreds synchro receiver

R. Lead X of hundreds synchro receiver

S. Lead X of tens synchro receiver

T. Lead Y of tens synchro receiver

U. Lead Y of units synchro receiver

V. Lead X of units synchro receiver

W. Leads Z and C of all distance synchro receivers

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X. +28V d-c servo amplifier power

Y. +28V d-c warning flag and thousands flag excitation

Z. Warning flag control (—)

a. Lead H of double-bar pointer synchro receiver

b. Leads Z and C of double-bar pointer synchro receiver

c. Thousands flag control (—)

d.
f.
g.
h. } SPARES

i. Lead H of compass card synchro control transformer

Note: Synchro leads H (hot), C (cold), X, Y, and Z are designated as defined in MIL-C-5824.

3.4.4.5 Synchros. The two-bearing pointers and each of the three distance indications (hundreds, tens, and units) shall be driven by a torque receiver synchro 26V-11TR4C, or an approved electrical equivalent. The synchro control transformer shall be a 26V-11CT-4d, or an approved electrical equivalent. All synchros shall be in accordance with MIL-STD-710.

3.4.4.5.1 Mechanical drive. Connections between the synchros and their corresponding dials or pointers shall be at a ratio of 1:1 and as direct as possible.

3.4.4.5.2 Voltage. The synchros shall operate from a voltage source of 26V, 400 ±20 Hz.

3.4.4.5.3 Electrical zero. Zero (North) indication of the compass card and of the distance indicators shall correspond to electrical zero of the synchros as defined in MIL-S-16892. Electrical zero of the bearing pointer synchros shall correspond to a bearing indication of 180° S when 0° N of the compass card is positioned under the lubber line.

3.4.5 Dial. The dial shall be marked as shown on figure 1. Unless otherwise specified, all visible portions of the dial shall be lusterless black, color No. 37038 of FED-STD-595. The dial shall be securely mounted in such a manner that it will not loosen or slip when the indicator is vibrated. If screws are used, they shall be so located that they will not interfere with the proper location of any dial marking.

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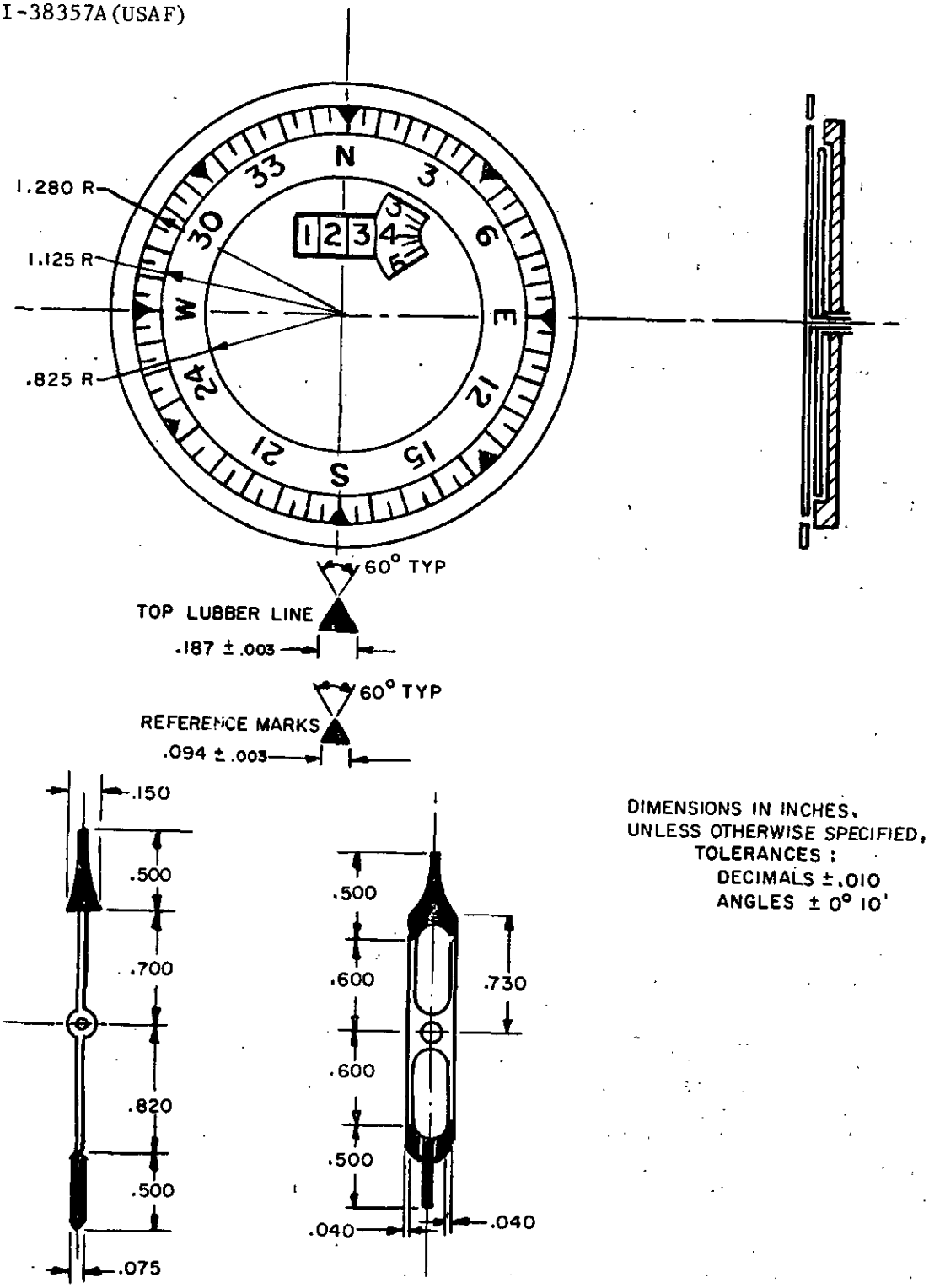


FIGURE 1. Dial and Pointers

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3.4.5.1 Dial and counter markings. The dial and counter markings shall be as shown on figure 1, and tolerances shall be as specified in table I. The form of the numerals and letters shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which each applies. Each numeral shall be so placed that the center of the area of the numeral is on the radial line joining the appropriate graduation and the center of the dial. When several numerals are used in one group, the space between the numbers shall not be less than 1/64 inch.

TABLE I. Markings

Marking	Height or Length ±0.010	Width of Line or Graduation ±0.003	Finish
Compass Card			
Numerals	0.219	0.027	Lusterless white,
Numbered grads	0.155	0.019	color No. 37875
Major grads	0.155	0.019	
Minor grads	0.100	0.016	
Zero index	0.187 ±0.003	---	
Other indices	0.094 ±0.003	---	
Letters	0.219	0.027	
Distance Indication Counters			
Unit numerals	0.170	0.021	
Other numerals	0.219	0.027	
Unit grads	0.120	0.020	
Half unit grads	0.080	0.013	
Background of OFF flag	---	---	
Background of dial	---	---	Lusterless black,
Letters OFF	0.094	0.016	color No. 37038

3.4.5.1.1 Pointers. The pointers of the indicator shall be firmly attached to the mechanism, and shall be readily adjustable. The shaded portion of the pointers shall be finished in lusterless white, color No. 37875 of FED-STD-595. The unshaded portion of the pointer shall be finished in lusterless black, color No. 37038 of FED-STD-595. The numerals on the pointers shall be 3/32 inch in height, finished in lusterless black, color No. 37038 of FED-STD-595, and in accordance with MS33558. The pointers shall conform to figure 1.

3.4.5.2 Graduations. The units distance indication shall be graduated in one-half-mile divisions as read against a stationary index. Compass and bearing divisions shall be every 5 degrees and heading information shall be indicated by means of a stationary lubber index as shown on figure 1.

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3.4.5.3 Compass card. The compass card shall be a stepped dial elevated toward the cover glass to establish the surface of the bearing pointer in the same plane as the surface of the compass card. The space between the pointed end of the double-bar pointer and the inner ends of the azimuth card large graduations shall be held to the absolute minimum required for clearance.

3.4.5.4 Visibility of dials. The numerals and letters and at least 1/16 inch of all compass dial graduations shall be visible from any point within the frustum of a cone whose side makes an angle of 30° with a perpendicular to the face of the instrument and whose smaller diameter coincides with the dial opening. Only one digit shall be visible on the thousands, hundreds, and tens distance indications, except during transfer when a sufficient portion of the going and coming numerals shall be visible for positive legibility. At least one digit shall be fully visible on the units indication at all times, with a sufficient portion of the adjacent coming or going numerals for easy legibility.

3.4.5.5 Rotation of indications. All distance indications shall so rotate that the numerals in the window travel upward for increasing values of distance.

3.4.5.6 Flag alarm. A flag alarm shall be provided and marked with the word OFF. The flag shall partially obscure the numerals and indicate OFF when the flag alarm circuit is not energized. The word OFF shall be in lusterless black letters approximately 3/32 inch high on a lusterless white background.

3.4.6 Cover glass. The cover glass shall be flat and free from discoloration, scratches, and striae that would interfere with the indicator. The cover glass shall be in accordance with MIL-G-81704 and its thickness shall not exceed 0.200 inch.

3.4.7 Lubrication. The indicator shall be lubricated in accordance with MIL-STD-838.

3.4.8 Soldering. Soldering shall be performed in accordance with MIL-STD-454.

3.5 Performance. The indicator shall meet the requirements of this specification when subjected to the following conditions:

a. Dielectric strength——A potential of 200V (rms) a-c commercial frequency

b. Vibration——Vibration at double amplitudes of 0.010 inch from 5 to 30 Hz; 0.005 inch from 52.3 Hz; and ±0.7g 52.3 to 500 Hz

c. High altitude-low temperature——50,000 feet at -55°C (-67°F)

d. Low temperature—— -65°C (-85°F)

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e. High temperature——Operating +70°C (+158°F)

f. High temperature exposure—— +70° (+158°F)

g. Humidity as encountered in any locale

h. Fungus growth as encountered in tropical climates

i. Salt spray——Exposure to salt sea atmosphere.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Screw threads. Screw threads shall be in accordance with MIL-S-7742.

3.8 Weight. The weight of the indicator shall not exceed 3.5 pounds.

3.9 Finish. Protective coatings and finishes that will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.9.1 Aluminum alloy parts. Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.9.2 Iron and steel parts. Where practicable, iron and steel parts shall be cadmium, chromium, or nickel plated in accordance with QQ-P-416, QQ-C-320, and QQ-N-290, respectively.

3.9.3 Indicator case. The body and mounting flanges of the indicator case specified in 3.4.3.1 shall be finished as follows:

3.9.3.1 Body. The body of the case shall be finished in a durable black finish.

3.9.3.2 Mounting flanges. The mounting flanges shall be finished in a durable lusterless black finish, color No. 37038 of FED-STD-595.

3.10 Nomenclature and nameplates. Unless otherwise specified, nomenclature assignment and nameplate approval for equipment shall be obtained in accordance with the requirements of MIL-STD-130.

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3.11 Workmanship. The indicator, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts, and tightness of assembly screws and bolts.

3.11.1 Controlled environments. The requirements of FED-STD-209 for clean room and work station environment shall be complied with.

3.11.2 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.11.3 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered, or excess solder; metal chips; and other foreign material removed during and after final assembly.

3.11.4 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practice. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.11.5 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests.

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4.3 Test conditions

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests shall be conducted in accordance with FED-STD-1.

4.3.2 Test position. Unless otherwise specified, the indicator shall be tested in a level position with the upper mounting holes in a horizontal line and the face vertical.

4.3.3 Tapping. Unless otherwise specified, the indicator may be gently tapped or vibrated before a test reading is taken.

4.3.4 Warmup. A 5-minute warmup time will be permitted prior to testing under low temperature conditions.

4.3.5 Tolerances. Tolerances indicated for room temperature shall prevail for environmental tests.

4.4 Preproduction tests

4.4.1 Test samples. The test samples shall consist of six indicators representative of the production equipment. Three indicators shall be subjected to all individual tests and the reliability test of 4.6.23 and the remaining three indicators shall be subjected to all the tests except reliability. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test report. When tests are conducted at a location other than the laboratory of the procuring activity, three copies of a test report in accordance with MIL-STD-831 shall be furnished that activity.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance tests. The quality conformance tests shall consist of the following:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. The individual tests shall consist of the following:

a. Examination of product

b. Burn-in

c. Dielectric strength test

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d. Accuracy tests

e. Position error test

f. Indicator speed test

g. Flag alarm test

h. Leakage

i. Damping test.

4.5.2 Sampling tests

4.5.2.1 Sampling plan. A minimum of three indicators shall be selected at random from each one hundred or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Individual tests

b. Vibration error test

c. High altitude——low temperature test

d. High temperature operation test

e. Power consumption test

f. Fogging test.

4.5.2.2 Rejection and retest. When an indicator selected from a production run fails to meet the specification, indicators still on hand or later produced shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.2.3 Individual tests may continue. For production reasons, individual tests may continue pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

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4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, marking, and dimensions.

4.6.2 Burn-in test. The indicator shall be operated for a minimum of 20 hours and the test data (operating hours and failures) shall be totaled and the results compared with the accept-reject criteria of test plan I of MIL-STD-781, with a minimum acceptable MTBF of 500 hours. (Extend the accept/reject lines as needed to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract or order is included. If the line of plotted test results indicates a reject situation, that is, if the plotted point crosses the reject line, then the procuring activity shall be notified immediately. The procuring activity reserves the right to stop acceptance of equipment at any time a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, workmanship, et cetera, so that the entire computation will show other than a reject decision.

4.6.3 Dielectric strength test. A potential of 200V (rms) alternating current at commercial frequency shall be applied between isolated pins and between pins and the case for a period of 5 seconds. There shall be no breakdown of insulation nor any other damage to the indicator as a result of this test.

4.6.4 Accuracy tests. The accuracy tests shall be conducted with three indicators connected in parallel and in accordance with figure 2. The synchro transmitters of figure 2 shall be Eclipse-Pioneer, Type AY201-1, or equivalent, selected or calibrated to have an electrical error of not more than 5 minutes of angle. The synchro transmitters shall be set to electrical zero (high null) in accordance with MIL-C-5824.

4.6.4.1 Compass card test. With the indicator connected as shown on figure 2, the compass card transmitter shall be rotated in a clockwise direction from null through 360° in increments of 30°. This test shall be repeated for rotation of the transmitting synchro in a counterclockwise direction. The compass card indication shall not differ from the synchro-transmitter setting by more than 1° at any test point.

4.6.4.2 Single-bar-bearing-pointer test. With the single-bar-bearing-pointer transmitter zeroed by the ARINC method, the single-bar-bearing-pointer shall be at the 180° index position of the indicator. The single-bar-bearing pointer shall be rotated from null through 360°, in both a clockwise and counterclockwise direction, in increments of 30°. The pointer shall not differ from the transmitter synchro setting by more than 1° at any test point.

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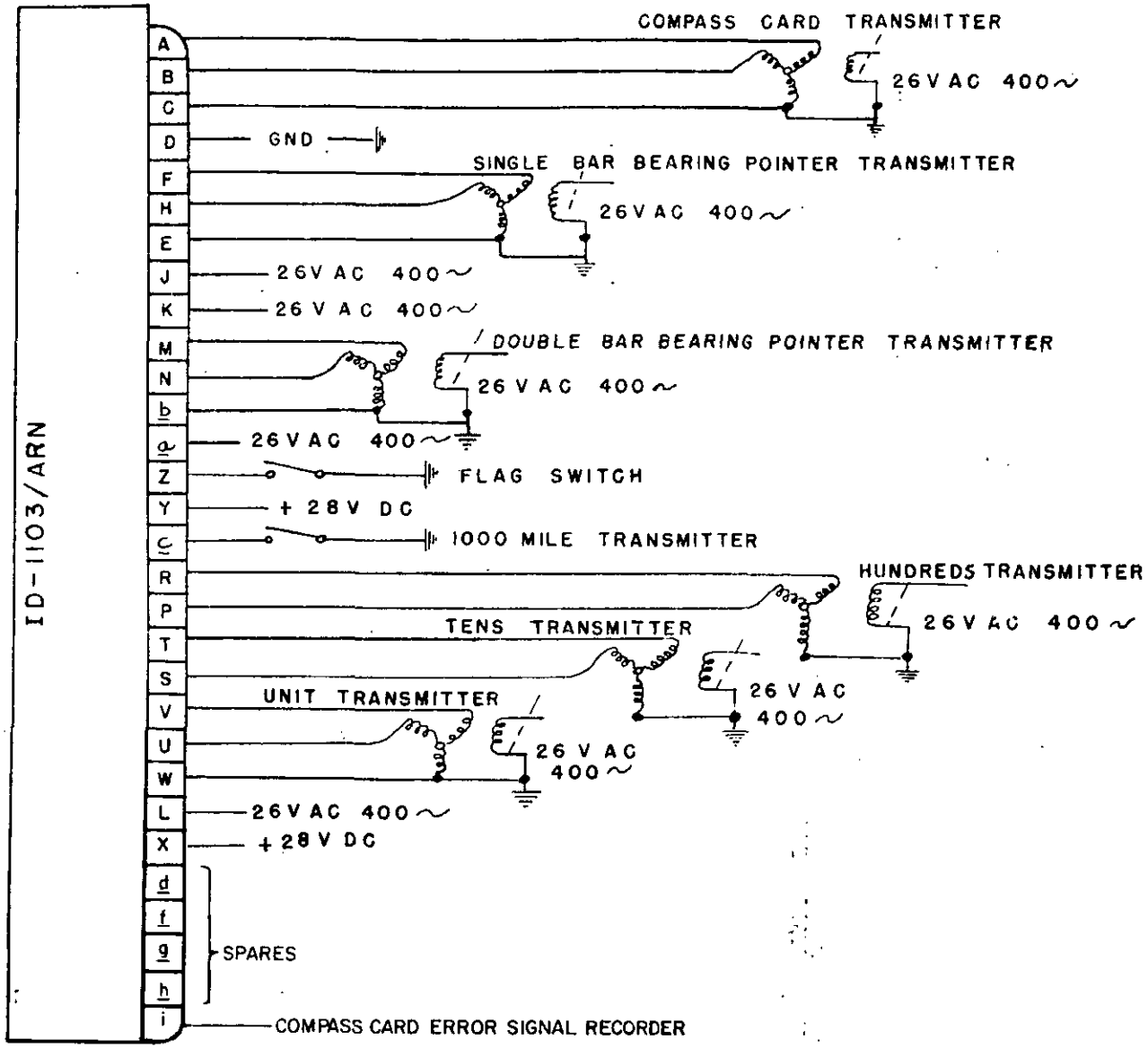


FIGURE 2. Test Circuit Wiring Diagram

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4.6.4.3 Double-bar-bearing-pointer test. The test specified in 4.6.4.2 shall be repeated, except the double-bar-bearing-pointer transmitter shall be positioned. The pointer shall not differ from the transmitter setting by more than 1° at any test point.

4.6.4.4 Distance presentation test. The test transmitter synchro associated with the units indication shall be rotated, starting at null, so that each digit of the units indication is exactly aligned with the index. The transmitter setting shall not differ from the proper multiple of 36° by more than 3.6°. The transmitting synchros for the remaining counters shall be rotated from null through 360° in increments of 36°. The correct digit shall be approximately centered in the viewing window for each position of the proper transmitting synchro.

4.6.5 Position error test. With the indicator connected as shown on figure 2, and with the indicator in the normal upright position, the indications for heading, bearing, and distance shall be noted. With the indicator held in any other position, the indications shall not differ from their original setting by more than the following:

Distance	0.1 Mile
Compass card	1 Degree
Bearing pointers	1 Degree

4.6.6 Damping test. With the instrument connected as shown on figure 2 and with no electrical power supplied, the rotors of the transmitter synchros shall be displaced 175° for the rotor position of their respective receiver synchros. The distance indication and the bearing pointers shall come to rest at their new positions in not more than 6 seconds.

4.6.7 Indicator speed test. With the indicator connected as shown on figure 2, all transmitter synchro rotors shall be displaced, in both the clockwise and counterclockwise directions, at the following rates and at a rate of 1.5 degrees/sec. The instrument indications shall be capable of following all input signals without damage, spinning, or excessive hunting. The hunting and jumping shall be within ±1/4 degree. The error signal from the compass card synchro control transformer shall be measured and shall not vary more than 0.1V peak-to-peak for any 0.5-second time frame. The indicator shall track the transmitter synchros within ±2 degrees.

Compass card	20 deg/sec
Bearing pointer	360 deg/sec
Distance presentation	180 deg/sec

4.6.8 Flag alarm test. With the instrument connected as shown on figure 2 and the switch open, the flag alarm shall partially obscure the numerals of the distance indication and indicate OFF. When the flag alarm switch is closed, the flag shall disappear revealing the numerals. The flag alarm operating current shall not exceed 5.0 milliamperes. The flag alarm and 1000-mile digit shall operate properly when the indicator is held in any position.

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4.6.8.1 Thousand-mile digit test. With the indicator connected as shown on figure 2 and with the switch open, the thousand-mile digit shall not be in view. When the thousand-mile switch is closed, the digit 1 shall center in the window.

4.6.9 Power consumption test. The indicator shall be connected as shown on figure 2 and the a-c and d-c power measured while the instrument is operating. The a-c power shall not exceed 45 va, and the d-c power shall not exceed 4.5w.

4.6.10 Leakage. The indicator shall be tested for leakage by means of a mass-spectrometer-type helium leak detector. The detected leak rate at a pressure differential of 12 to 14.7 psi shall not exceed 0.1 micron cubic foot per hour.

4.6.11 Vibration error test. With the indicator connected as shown on figure 2 and energized, it shall be so mounted on the vibration stand that a line joining the centers of the two lower mounting holes is in a horizontal plane and the plane of the mounting lugs in a vertical position. The indicator shall be subjected to vibration at frequencies from 5 to 500 Hz at the following maximum double amplitudes or corresponding maximum vibration accelerations; 0.10 inch from 5 to 30 Hz, 0.005 inch from 30 to 53.2 Hz, and ±0.7g vibratory accelerations from 52.3 to 500 Hz. The indicator shall be vibrated in each of three mutually perpendicular planes with the frequency being varied uniformly from 5 to 500 Hz and returned during a period of approximately 15 minutes.

4.6.11.1 Pointer vibration. The maximum amplitude of the pointers' vibration shall not exceed 1/64 inch.

4.6.11.2 Pointer accuracy. The bearing reading shall not differ at any time by more than 1/2° from the reading obtained when the indicator is at rest.

4.6.11.3 Distance accuracy. The distance indication shall be legible and accurate to within ±0.1 mile.

4.6.11.4 Compass card accuracy. The compass indication shall not differ at any time by more than 1/2° from the indication when the indicator is at rest.

4.6.12 High altitude-low temperature test. The indicator shall be subjected to an altitude-temperature test in accordance with MIL-STD-810, method 504, procedure I as specified for class 1 equipment with continuous operation. Step 5 and steps 8 through 14 will be omitted. During steps 2 and 6, the tests specified in 4.6.4, 4.6.7, and 4.6.8 shall be performed.

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4.6.13 Low temperature exposure test. The indicator shall be subjected to a temperature of -65°C (-85°F) for a period of 48 hours. The temperature shall then be raised to -55°C (-67°F) and maintained for a period of 25 hours. While at this temperature, the indicator shall be subjected to the following tests:

a. Accuracy tests

b. Indicator speed test

c. Flag alarm test.

4.6.14 High temperature operation test. The indicator shall be subjected to a temperature of +70°C (+158°F) for a period of 4 hours. While at this temperature the indicator shall be subjected to the following tests:

a. Accuracy tests

b. Indicator speed test

c. Flag alarm test.

4.6.15 High temperature exposure test. The indicator shall be subjected to a temperature of +70°C (+158°F) for a period of 24 hours. After the indicator has returned to room temperature, it shall be subjected to the following tests:

a. Accuracy tests

b. Indicator speed test

c. Flag alarm test.

4.6.16 Vibration failure test. The indicator shall withstand, without damage, vibration at frequencies from 5 to 500 Hz at the following maximum double amplitude of corresponding maximum vibratory accelerations: 0.020 inch from 5 to 50 Hz, 0.006 inch from 50 to 80.7 Hz, and ±2g from 80.7 to 500 Hz. The indicator shall be vibrated for 1 hour in each of the three mutually perpendicular planes with the frequency being varied uniformly from 5 to 500 Hz and returned during a period of approximately 15 minutes. Upon completion of the vibration test, the following tests shall be performed:

a. Accuracy tests

b. Indicator speed test

c. Flag alarm test.

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4.6.17 Humidity test. The indicator shall be tested in accordance with method 507, MIL-STD-810. At the conclusion of the test and within 1 hour after having been removed from the test chamber, the indicator shall be subjected to the dielectric strength test and to the following tests:

a. Accuracy tests

b. Indicator speed test

c. Flag alarm test.

4.6.18 Fungus resistance test. The indicator shall be tested in accordance with method 508, procedure 1, MIL-STD-810, except that the exposure period shall be 14 days. There shall be no evidence of fungus growth on the indicator as a result of this test.

4.6.19 Salt spray test. The indicator shall be tested in accordance with method 509 of MIL-STD-810. The indicator shall be subjected to the following tests:

a. Accuracy tests

b. Indicator speed test

c. Flag alarm test.

4.6.20 Fogging test. The indicator shall be energized and placed in a $71^{\circ} \pm 2^{\circ}C$ controlled ambient temperature for a minimum period of 1 hour. After this period, and while still at this temperature, an ice cube shall be rubbed on the cover glass for a period of from 1 to 2 minutes. The glass shall be wiped dry (do not use compressed air), and the indicator shall be inspected for evidence of water or oil fog inside the cover glass. Evidence of fogging shall be cause for rejection.

4.6.21 Mounting lugs. The indicator case shall be mounted face downward on the movable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for 1 minute to each lug in a direction toward the front of the case. The lugs shall withstand the applied load without fracture, and there shall be no damage to any part of the indicator.

4.6.22 Electromagnetic interference. The indicator shall be properly connected and tested for conducted and radiated electromagnetic interference in accordance with MIL-STD-462 for class A1 equipment.

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4.6.23 Reliability test. Three indicators shall be subjected to a reliability test in accordance with MIL-STD-781, test plan III, test level F, except vibration shall be 1g. Reporting shall be in accordance with MIL-STD-781. During each operational period, heating cycle only, with the indicator connected as shown on figure 2, the bearing pointers, compass card, and distance presentation shall be cycled at a rate of 20 degrees per second. The direction of rotation shall be reversed every 2 minutes. The flag alarm and 1,000-mile switch shall also be operated every 2 minutes. At approximately 200 ±10 hour intervals and at the completion of the reliability test, the indicator shall be subjected to and shall pass all the individual tests.

4.6.23.1 Accept/reject criteria. The accept/reject criteria shall be in accordance with MIL-STD-781, test plan III.

4.6.24 Longevity. Upon completion of the testing specified under 4.6.23, testing of one of the indicators shall be continued until a total of 2,500 operating hours are accrued on the indicator. Testing shall be in accordance with MIL-STD-781, test plan XXVIII, except the environment shall be room ambient with no vibration applied. Power-off periods shall be 5 minutes for every 2 operating hours. Performance checks and test conditions shall be as specified in 4.6.23. Each failure shall be recorded and the equipment repaired prior to resuming this test.

4.6.25 Maintainability demonstration. A maintainability demonstration test shall be conducted in accordance with MIL-STD-471, appendix B, test method I, test plan A plus B. The indicator shall meet the requirement of 3.4.2.2.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Indicators shall be preserved and packaged in accordance with MIL-P-116, method 1C1. QUP shall be one. Each wrapped, cushioned, and bagged indicator shall be packaged in a unit container conforming to PPP-B-636, weather-resistant class. The unit container shall be lined with 2-inch-thick cushioning conforming to MIL-P-26514, type I, class 2, 1.45 density.

5.1.2 Level C. Preservation and packaging shall be the minimum necessary to afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

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5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in containers conforming to PPP-B-636, weather-resistant class, or PPP-B-640, class 2. Gross weight shall be limited to 200 pounds for the PPP-B-636 containers.

5.2.2 Level B. Packing shall be in accordance with 5.2.1, except that containers shall conform to PPP-B-636, domestic class, and PPP-B-640, class 1.

5.2.3 Level C. Packages which require over-packing for acceptance by carrier shall be packed in shipping containers in a manner which will insure safe delivery at the lowest transportation and packing charges combined. Containers shall conform to Uniform Freight Classification Rules and Regulations or other rules and regulations applicable to the mode of transportation.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

INDICATOR, BEARING, DISTANCE, HEADING ID-1103/ARN.

6. NOTES

6.1 Intended use. The ID-1103/ARN indicator is intended for use in aircraft to give bearing, distance, and heading information.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Levels of packaging and packing required (see 5.1 and 5.2).

6.3 International standardization agreements. Certain provisions of 3.4.5.1 of this specification are the subject of international standardization agreements ASSCC 10/6 and STANAG 3329. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.4 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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